

## Sarna Ma'am's Part

### Peripherals and Interfaces

2019-20

4-C (3)

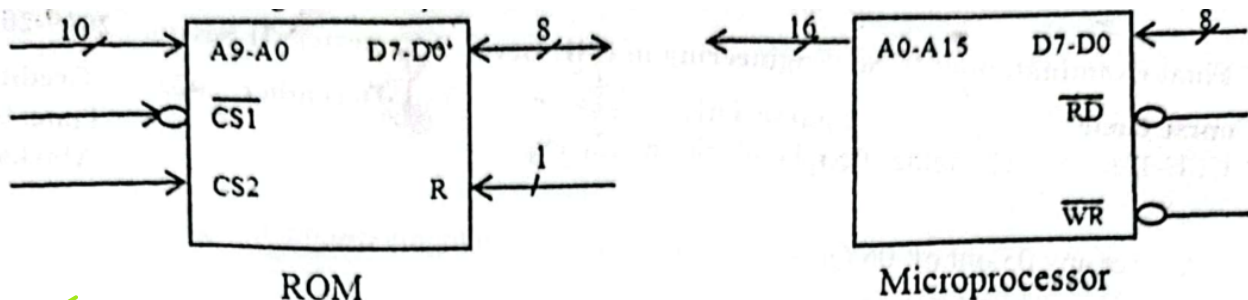
Convert  $-1313.3125_{10}$  to IEEE 32-bit floating point format.

4-D (2)

Mention the main functions of an interface.

5-A (5)

Assume the following ROM chip and the Microprocessor:



Connect the microprocessor to the ROM to obtain the memory map:  $0000_{16}$  through  $03FF_{16}$ .

Use only the signals shown. Draw a neat logic diagram and analyze the memory map.

2018-19

4-B (4)

Interpret  $EFE_{16}$  as unsigned, signed, and floating point numbers that a microprocessor can understand.

5-D (2)

Make a comparison between buffers and spoolers.

6-A (5)

Suppose, a processor is having 16 bits address line and it is byte addressable. Interface  $4096 \times 8$  RAM with the processor.

- What is the memory capacity of the processor?
- How many address lines are required for a RAM?
- How many RAM chips are required?
- What will be the size of address decoder?

## Rafiquzzaman - 1st Chapter

2019-20

2018-19

5-C (3)

The microcomputer communicates with an external device via one or more registers called I/O ports using programmed I/O. Illustrate the functions of two of them with brief description.

## Interfacing Keyboard

2019-20

5-B (4)

When you press a key on your computer, you are activating a switch. There are many different ways of making these switches. Describe any four of them.

5-D (2)

What is key debouncing? Why is it needed?

2018-19

6-B (4)

Show the software keyboard interfacing process through a flowchart. Describe each step for a  $4 \times 4$  keyboard.

## Image Representations

2019-20

6-A (5)

i. If we want to cut a  $512 \times 512$  sub-image out from the center of an  $800 \times 600$  image, what are the coordinates of the pixel in the large image that is at the lower left corner of the small image?

ii. Find the CMY coordinates of a color at (0.2, 1, 0.5) in the RGB space.

iii. If we use direct coding of RGB values with 2 bits per primary color, how many possible colors do we have for each pixel?

iv. What is the pitch of a color CRT?

v. Why do many color printers use black pigment?

2018-19

4-A (5)

a. What do you mean by true color representation? Does it represent any disadvantages? How to handle the problems?

b. If we use 5 bits each for red, and blue and 6 bits for green in direct coding method, how many possible simultaneous colors do we have?

## Display Basics

2019-20

5-C (3)

Explain in detail the working principle of an LCD monitor and how its components interact to produce images on the screen?

6-D (2)

Provide a detailed comparison of Random Scan Displays and Raster Scan Displays, focussing on their operational mechanisms, image rendering techniques, applications, and overall performance characteristics.

2018-19

6-C (3)

Serialize the steps associated in turning on or off a pixel in LCD monitor.

## 8255 PPI

2019-20

4-A (5)

Show the pin configuration of 8255 PPI and separately describe the operation performed by each unit.

6-B (4)

How can we assign input and output port of 8255?

2018-19

4-D (2)

Obtain the control word when the ports of 8255A are to be used in mode 0 with port A as output port and port B as input port and port C as output port.

## Printer

2019-20

4-B (4)

Illustrate the key components of a laser printer and provide a brief description of the function each component performs.

2018-19

4-C (3)

What are the common problems with scanner? Share some tips to troubleshoot the basic scanner related problems.

6-D (2)

An interface incorporates several components including its communication type as well as the interface software. List out eight major communication types in case of printer.

## 8279 Programmable Keyboard Display Controller

2019-20

2018-19

5-B (4)

Categorize and explain the operational modes of 8279. How many ways the keyboard is interfaced with the CPU?

## USB (~~not~~ in syllabus)

2019-20

6-C (3)

In USB communication, a single transaction involves the transmission of up to three packets. Mention the names of these packets and describe the specific services each one provides during the communication process.

2018-19

5-A (5)

How do the USB peripherals communicate with microprocessor? Give the pin-out for USB type C 3.0.

Arpita Ma'am's Part

Not Sure

2019-20

2-C (4)

Imagine you are designing a real-time system for a futuristic smart city where various subsystems need to interact seamlessly to ensure optimal performance. In this context, how would you incorporate both periodic and aperiodic task models to manage different types of operations?

3-A (3)

Define nonlinearity error. How it is different than Hysteresis error?

3-B (3)

What are the key components and characteristics of a digital ecosystem, and how do these elements interact to create value within industries such as finance, healthcare, and e-commerce? Additionally, discuss the challenges and opportunities associated with managing and sustaining a digital ecosystem in a rapidly changing technological environment.

2018-19

1-C (4)

Write down the difference between Embedded and Real Time Systems.

2-A (4)

What is the periodic and aperiodic task model in real-time system? Explain with example.

3-A (3)

Explain the characteristics of RTS.

Internet of Things (IoT)

2019-20

1-A (4)

What are the different components of IoT? How does the Internet of Things (IoT) affect our everyday lives? [2018-19 : 1-A]

2-A (3)

What are the different types of IoT security measures, and what are some effective strategies or best practices for securing IoT devices to protect them from potential threats and vulnerabilities?

2018-19

1-B (6)

What is communication model in IoT? Explain different types of communication model in IoT with diagram and example.

2-D (3)

Briefly explain the most 6 communication protocols used by IoT.

3-D (3)

What are the types IoT security? How to secure IoT devices?

## Human Computer Interaction (HCI)

*[mentioned the idea, but not in syllabus, should check it out]*

2019-20

1-C (5)

What is Human Computer Interaction? Briefly explain most important components of HCI. [2018-19 : 3-C]

2-D (3)

Write down Norman's Seven Principles.

2018-19

2-C (3)

How many principles are there in Norman design principles? Write down Norman's design principles for IoT.

## Sensors and Signal Processing

2019-20

1-B (5)

Explain the structure of a potentiometer sensor used for measurement of linear displacement with electric circuit diagram and proper equation. List the applications of potentiometer sensor in/around your home and office/university.

2-B (4)

How does a capacitive sensor operate to measure linear displacements, and what are the typical applications where such non-contact sensors are preferred over other types of displacement sensors? Additionally, discuss the range of displacements that capacitive sensors are capable of measuring and factors that affect their accuracy. Explain all with appropriate diagram.

3-C (5)

What is a LVDT? How does LVDT work step by step? Explain with LVDT construction diagram.

3-D (3)

What are the definitions and characteristics of active and passive transducers, and how do they differ from each other? Additionally, what are the key distinctions between sensors and transducers in term of their fucntions and applications?

2018-19

2-B (4)

What is RVDT? How does RVDT work step by step? Explain with RVDT construction diagram.

**3-B (3)**

Write down the difference between Sensors and Transducer.

## **WSN Taxonomy on Communication & Power**

**2019-20**

**2018-19**

## **Personal Area Network (PAN) Protocol – Medium Access Control (MAC)**

**2019-20**

**2018-19**

## **Synchronous MAC Protocols for Wireless Sensor Networks**

**2019-20**

**2018-19**